Biology Chapter: A City Garden Ecosystem

A short, fictional field guide for testing document upload, semantic search, and source-grounded answers. It describes one community garden across three connected topics.

1. What makes an ecosystem?

An ecosystem is a community of living organisms interacting with one another and with nonliving parts of their environment. In the Riverside Community Garden, living components include tomato plants, clover, bees, earthworms, spiders, and people. Nonliving components include sunlight, air, water, soil minerals, and temperature. A change in one component can affect many others.

Plants are producers: they capture sunlight and store some of its energy as chemical energy through photosynthesis. Herbivores such as caterpillars and grasshoppers eat plant material. Spiders and small birds act as consumers by eating insects. Decomposers, including fungi and bacteria, break down dead leaves and return mineral nutrients to the soil. Matter is recycled through the ecosystem, while energy flows from sunlight through organisms and eventually leaves as heat.

A food chain is one simple path for energy, such as **sunlight ? clover ? grasshopper ? spider ? bird**. A food web combines many connected feeding paths. A bird may eat spiders and caterpillars; a spider may eat several kinds of insect. Because organisms often have more than one food source, a food web usually describes a garden more accurately than a single chain.

Why are there fewer hawks than grasshoppers in an energy pyramid?

Only a fraction of the energy stored at one feeding level becomes new biomass available to the next. Much of the energy is used for movement and life processes or released as heat. A useful classroom estimate is the 10 percent rule: roughly one tenth may pass to the next trophic level, although real ecosystems vary. Therefore, an area can support many plants, fewer grasshoppers, and still fewer insect-eating birds or hawks. The rule describes energy transfer, not a fixed count of every species.

Garden observation: when the gardeners left a strip of clover beside the vegetable beds, more bees visited flowers and more spiders were seen among the leaves. This observation suggests possible connections, but by itself does not prove that clover caused every change. Repeated counts and comparison plots would provide stronger evidence.

2. Nutrient Cycles and Garden Soil

Unlike energy, atoms such as carbon and nitrogen are reused. Nutrient cycles move these atoms among organisms, soil, water, and the atmosphere. Gardeners can influence these cycles through composting, watering, planting, and soil disturbance.

The carbon cycle

Plants take in carbon dioxide from the air during photosynthesis and use carbon to build sugars and plant tissues. Animals obtain carbon by eating plants or other animals. Respiration returns some carbon dioxide to the air. When leaves and organisms die, decomposers break down their material; some carbon enters the soil, while some returns to the atmosphere. Composting recycles carbon-rich plant matter, but compost is not an instant substitute for balanced soil nutrients.

The nitrogen cycle

Most nitrogen in the atmosphere is nitrogen gas, which plants cannot use directly. Certain soil bacteria convert nitrogen gas into compounds that can enter the food web. Other bacteria transform nitrogen compounds as they decompose organic matter or as oxygen conditions change. Plant roots absorb usable nitrogen compounds from soil water; animals obtain nitrogen by eating plants or other animals. Decomposition returns nitrogen to soil. Legumes such as beans can partner with nitrogen-fixing bacteria in root nodules, adding biologically available nitrogen to the system.

How does nitrogen become available to plants?

Nitrogen-fixing bacteria convert atmospheric nitrogen into usable compounds, and soil processes also release nitrogen from decomposing organic matter. Plant roots then absorb usable nitrogen compounds, including nitrate and ammonium, from soil water. Beans can support nitrogen fixation through bacteria living in their root nodules; simply planting any crop does not fix nitrogen.

Water and soil

Rain and irrigation dissolve mineral nutrients and carry them through soil pores. Plant roots take up water; leaves release water vapor through transpiration. Some water evaporates from soil, and some moves into groundwater or drains away. Mulch can reduce evaporation and slow runoff. Overwatering can wash soluble nutrients out of a bed, while compacted soil can reduce air spaces needed by roots and soil organisms. A gardener can compare mulched and bare plots, keeping other conditions similar, to investigate water loss.

Practical example: adding mature compost to a vegetable bed can improve soil structure and return nutrients from decomposed organic matter. The result depends on the compost, soil, and crop. Testing soil before adding fertilizer helps avoid excess nutrients, which can wash into streams and encourage unwanted algal growth.

3. Biodiversity, Pollinators, and Restoration

Biodiversity includes variation among genes, species, and ecosystems. A garden with different flowering plants, nesting places, and soil habitats can support more kinds of organisms than a single-crop plot. Biodiversity does not guarantee that every ecosystem is healthy, but it can provide ecological functions and options when conditions change.

Why pollinators matter

Bees, butterflies, moths, flies, and some beetles move pollen between flowers as they feed. For plants that depend on animal pollination, pollen transfer can enable fertilization and seed production. Not every crop needs an animal pollinator, and pollinator visits alone do not guarantee a harvest: water, light, temperature, and plant health matter too. Flowering plants that bloom at different times can provide food across more of the growing season.

A habitat strip can offer nectar, pollen, shelter, and nesting sites. Native plants suited to local conditions are often useful, but a good planting plan should consider local guidance, bloom times, pesticide exposure, and whether plants could spread invasively. Reducing broad-spectrum pesticide use and leaving some undisturbed ground can also make a garden more welcoming to beneficial insects.

A small restoration study

Suppose a garden team wants to know whether a flower strip helps pollinators. They establish a flowering strip along one bed and keep a similar bed without the strip as a comparison. For four weeks, they count pollinator visits to marked flowers at the same time of day, using the same observation period and avoiding rainy or unusually windy sessions. They also record which flower species are blooming. Repeating observations on several days reduces the influence of one unusual visit.

If the flower-strip bed receives more visits, the result is evidence of an association, not definitive proof of causation. Differences in shade, nearby nests, or plant density could also matter. A stronger study repeats the comparison across multiple beds and seasons. The team should report both the observation method and limitations rather than claiming that one garden proves a universal rule.

What a flower strip can?and cannot?do

A flower strip can provide food and habitat that may increase pollinator activity near the garden. It cannot by itself guarantee crop pollination, reverse every population decline, or replace larger connected habitats. The most useful actions combine diverse flowering plants, nesting resources, reduced pesticide exposure, and careful monitoring. The best next step is to compare observations over time and adapt the planting to local species and conditions.

Key takeaway: ecosystem questions are connected. Flower diversity can affect pollinator food; pollinators can affect seed formation in some plants; soil organisms recycle nutrients; and energy availability influences how many organisms each feeding level can support. Use the specific evidence in each passage when answering questions, and distinguish a measured result from a suggested explanation.